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Review

Recent advances in sensor fault diagnosis: A review

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ABSTRACT

As an essential component of data acquisition systems, sensors have been widely used, especially in industrial and agricultural sectors. However, sensors are also prone to faults due to their harsh working environment. Therefore, the early identification of sensor faults is critical for making corrective actions to mitigate the impact. This paper provides a comprehensive review on the contemporary fault diagnosis techniques and helps researchers and practitioners to understand the current state of the art development in this emerging field. The paper introduces the common fault types and causes in sensors, and different types' methods for fault diagnosis used in industry and agriculture sectors. It discusses the advantages and disadvantages of these methods, highlights the current challenges, and offers recommendations for future research directions.

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1. Introduction

Sensors are widely used for acquiring information and signals, especially in environmental monitoring [1–3], aquaculture system monitoring [4], disease detection [5,6], machinery monitoring system [7] and agricultural monitoring [8,9]. For example, in agriculture sector, sensors are used for collecting data on soil moisture, temperature and nutrients for prediction of crop health and production quality [10]; for fruit and vegetable picking using machines and robots [11]; and for environments monitoring [12]. Sensors provide the primary support for precision agriculture and intelligent control. Most sensors are placed in harsh environments (such as high temperatures, underwater), making them prone to fouling and damages, which will affect the accuracy, stability and reliability of the system [13].

In the event of sensor failure, the performance of the system may be degraded, which can lead to catastrophic consequences [14,15]. The early identification of faults in sensors is the most important aspect of fault detection to ensure data accuracy and reliability [11]. Fault diagnosis of sensor is an important approach to improve the reliability of the entire fault diagnosis system. In complex environment, there are various causes for sensor performance degradation [16]. For early sensor fault diagnosis, the main solution is to use multiple sensors to measure the same parameter at the same time and treat the deviation in the measurement results as fault. This method is usually effective, but the excessive use of sensors will not only increase the cost, but also make the system more complex. In recent years, automated tools such as knowledge based systems, neural networks and fuzzy logic-based systems are used in fault detection decisions.

This paper presents a review on recent progress in sensor fault diagnosis (our study does not include research on soft sensors). As shown in Fig. 1, the review follows the fault diagnosis process of sensors. This paper first lists the sensor faults classification and then discusses the common fault diagnosis techniques, focusing on recently proposed methods, related technologies and their advantages. The conclusion section highlights the key findings and suggestions for future research.

2. Types of sensors faults

A failure of a sensor means a partial or complete loss of functionality. The continuing use of a fault sensor without corrective measures will cause serious problems to the sensor based information systems [17]. During the operation of the sensor [18], the cause of the fault could be external or internal. External causes include environmental factors such as temperature, humidity, pollution, corrosion, etc. Internal causes include component wear and overheating, etc. [17]. When Tolle et al. [19] deployed sensors in the field, they found that only 49 % of the final data measured by the sensors were valuable because of the complex working environment.

For the classification of sensor failures, Ni et al. [20] classified sensor faults based on sensor data characteristics in their most detailed article on sensor fault classification. In addition, they provided a list of common sensor fault data with a very comprehensive analysis of fault types. They also listed the duration and impact of the fault, which provided valuable guidelines for future research. The common types of faults are discussed below.

Based on [21], we classify types of sensor faults into two categories: incipient (slowly developing) failure and abrupt (sudden) failure. While defining these faults, we summarize the characteristics of each specific failure.

2.1. Incipient failure

The “Incipient Failure” indicates a sensor system in abnormal or unstable working status, although the sensor can continue working, the data is not correct [22]. The error parameters may be small and change slowly, but the degree of errors gradually increases over a long period. As a result, the incipient faults will further develop into serious failures. This type of faults is difficult to detect and diagnose in the early stage. Only by correctly classifying and modeling incipient faults, it can be effectively detected and diagnosed.

2.1.1. Sensor Bias

Bias is one of the most common faults in sensors, and the data value of the sensor is replaced by a constant value [23]. For temperature sensors, temperature drift can occur with the increase of service time, and the physical characteristics of the sensor will change. This fault is usually caused by the existence of bias voltage or bias current.

Bias occur as a result of external interference, the sensor may loss its accuracy [24]. Ramanathan et al. [25] classified deviation faults as short-circuit faults. Ni et al. [20] provided a language description for this fault, and they used features to achieve corresponding fault detection. Sheng et al. [26] studied fault detection for deviation from normal value in sensor network deployment. Joshi et al. [27] gave a mathematical definition of the bias fault. Recently, He et al. [28] dealt with sensor bias faults and proposed a fault estimator. Wu et al. [29] showed that the sensor’s bias fault will affect the stability of the entire system, so they designed a diagnosis system for identifying the sensor bias faults.

In order to detect and identify deviation faults, it is necessary to establish a correct time domain or a spatial model. The unstable change of ambient temperature will cause the instability of bias voltage, which will lead to the sensor bias fault. This kind of failure is not easy to find, and can cause many unexpected problems due to the failure.

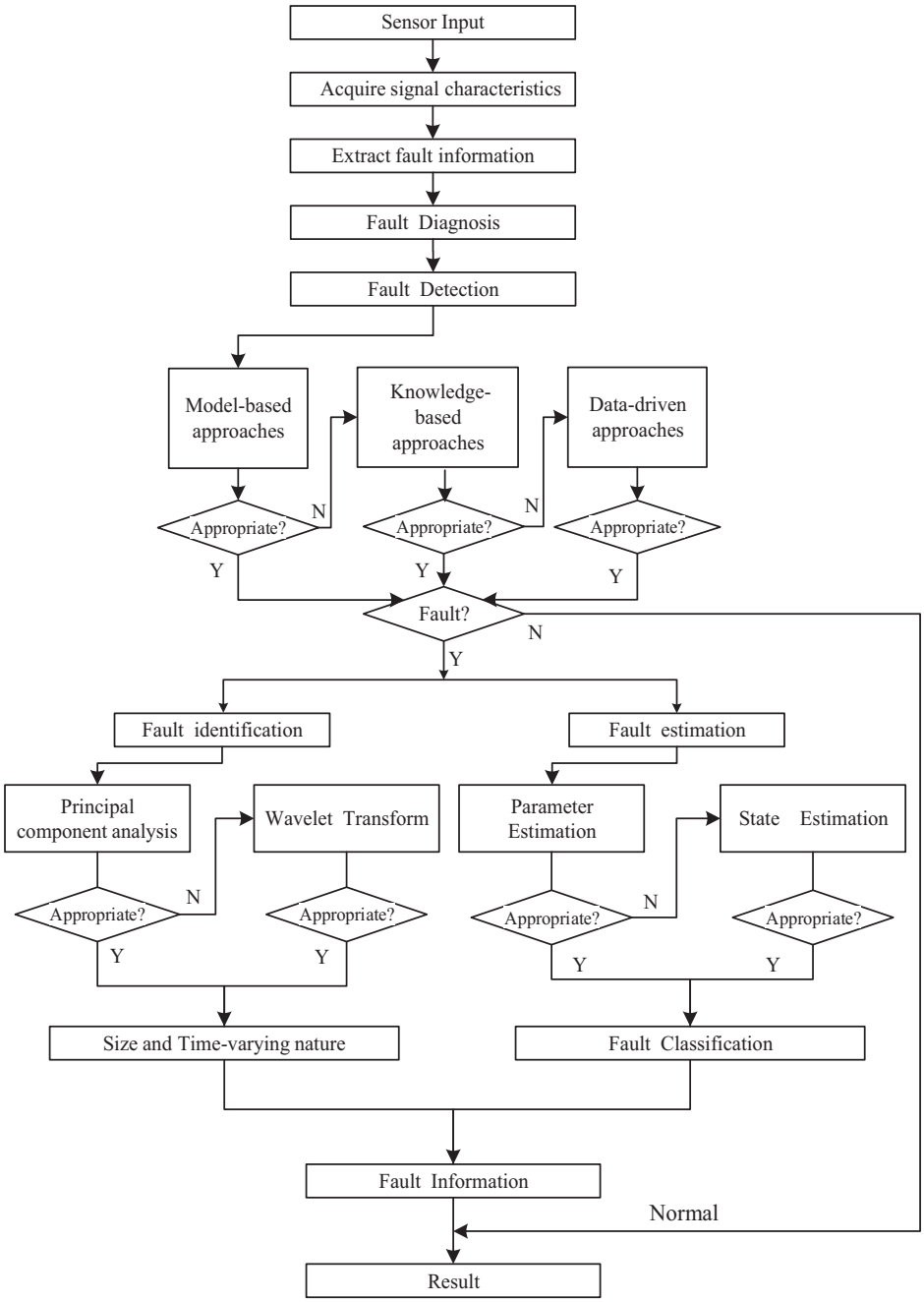

Fig. 1. A process framework for sensor fault diagnosis.

2.1.2. Sensor drift

Sensor drift is a time-varying offset, which means that the sensor performance deviates from the original calibration formula, and the offset or gain parameter changes with time [23]. Drift is normally small, with the data of failure changes very slowly. Sensor drift is not easy to be noticed in the early days, so the early detection and correction is very important.

The measured value of the sensor with drift fault changes with time. For example, a light sensor will only be able to read 60 % of the normal brightness because of dust or other obstacles gathered on the sensor. Another example is that the bridge circuit parameters of the pressure sensor are asymmetric. Since the temperature coefficient of the bridge circuit is different to some extent, the output of the bridge circuit is unbalanced. The sensor will produce zero drift. Lee et al. [30] found that the change of humidity would lead to changes of the measurement capacitance between the micro-cantilever and the substrate, leading to temperature drift of the sensor. In fact, many soil sensors are prone to drift faults [31]. Ramanathan et al. [32] deployed sensors in the soil, and they found that most of the sensors in the experiment could not reflect the actual measurement due to drift. They concluded that the way to capture the drift of the sensor is to occasionally recalibrate the sensor and simulate the drift between calibration events. The calibration itself requires a lot of work. It is even more time consuming and troublesome for buried sensors because the sensors need to be removed from the ground, calibrated and put back afterwards.

Drift may occur from the sensors' own structural parameters or the surrounding environment.

2.1.3. Faults in the sensor gain

The fault of gain refers to that the signal output of a faulty sensor is not zero [33], but the rate of change of the data being measured over an extended time period does not meet expectations. This can be described mathematically by stating that whenever a phenomenon changes by any amount,△, the sensor reports a change of B*△, where B is a positive real value.

Ni et al. [20] provided a simple definition of gain in their paper. Anthony et al. [34] proposed an algorithm that allows the gain error to be modeled, and allows for the calibration of the gain of a sensor. In order to facilitate the detection of such faults, Wang et al. [35] proposed a division of faults of gain into conditional fault identification and conditional fault detection. In addition, Balzano [25] proposed an algorithm for gain fault recognition based on singular value decomposition and standard least square technology. This method not only allows for a theoretical description of the gain calibration problem, but also proves that the theorem allows for the complete restoration of gain, without noise or other errors. Liu et al. [36] provided a model of the gain error, and calibrated the fault of gain in sensors in order to address this type of faults.

2.2. Abrupt failure

"Abrupt Failure" mostly refers to the sensor systems triggered by physical damage, which causes sensors to stop working suddenly [21]. When the sensor abruptly or completely fails, a sudden fault parameter that is easily identifiable in a short time will be generated. Since these mutations are obvious, such faults are usually detected by simple logical reasoning, and thus, are easy to be detected and eliminated.

2.2.1. Sensor noise

The sensors' operating environment can be complex and susceptible to noise from internal, hardware noise, and environmental noise. Internal noise mainly comes from the noise of the sensor itself and circuit elements, such as amplifier noise. External noise is caused by human or external interference outside the sensor circuit. Although noise is common in sensor data, unusually high noise can cause problems in sensor signals [21].

People noticed sensor data could be easily affected by some error sources, for example, external source of noise, hardware and environmental noise. Combined with the prior knowledge of the actual sensor reading and the characteristics of the sensor noise, a method is proposed to reduce the sensor noise on the premise of observing the reading noise. Sharma et al. [35] proposed a mathematical definition of noise, and noise faults can be detected by means of learning-based method. An example provided by Ramanathan et al. [25] illustrates a possible consequence of the low battery sensor, which the low battery sensor reading causes noise failure and the sensor does not work properly.

In general, there are many reasons for noise, which is one of the common faults. The signal of the sensor is easily covered by the noise signal, which affects the accuracy of the sensor data. Effectively restraining and reducing the noise is the only way to make better use of the sensor.

2.2.2. Short circuit and open circuit

Sensors can encounter hardware failures. For example, a sensor with poor contact and connection is often short-circuited, and disconnection of the signal line may trigger open-circuit faults in the sensor. In addition, some special chemical sensors may be subject to ion interference, resulting in inaccurate data when deployed in complex environments [20,37]. These issues are part of a hardware failure of the sensor, which are not difficult to notice when they appear.

If the internal connection of the sensor is disconnected, it will cause the sensor's bias voltage to be equal to the supply voltage. Once the cable or the internal connection is short-circuited, the bias voltage of the sensor will drop close to zero. In addition, when the temperature sensor comes in direct contact with water, the sensor hardware will be damaged [38–40]. Szewczyk et al. [41] used temperature and humidity sensors to directly contact with water, resulting in a short circuit between the power terminals, where the sensor reading was anomalous. Similarly, Ni et al. [20] attributed the problem of sensor failure to the sensor being submerged in water for a prolonged period, thermal resistance caused the sensor to be poorly connected, resulting in the sensors eventually loss of its ability to function. Silva [42] showed in their research that sensor failures often originate from hardware problems, such as connection failure.

Connection and hardware failures are typically indicated by unusually high or low sensor readings. The hardware failure elimination usually requires a suitable hardware failure location, replacement, or sensor repair.

2.2.3. Sensor random faults

Due to the complex layout environment, the sensor often experiences random failures. For example, when the sensor is contaminated, the sensor's bridge circuit will be corroded, which will cause the sensor to malfunction. When the sensor environment exceeds the requirements of the sensor, the data deviation will cause the fault, and these faults will disappear when the environment returns to normal [43].

Szewczyk [41] conducted a four-month deployment on the island. These sensors had experienced different weather conditions (such as heavy fog, rain and floods) during the test. They not only measured each node of sensors, but also analyzed the environment and the node health data. For the test of humidity sensors, rain caused sensors producing a large abnormal value for the reading. Ramanathan et al. [44] conducted a sensor failure simulation test by burying chemical sensors in soil to monitor chemical concentration. The authors showed that the soil environment caused the wrong data. Balzano et al. [39] mentioned that during the deployment of chemical sensors, the concentration reported by a sensor of chloride ion exceeded the total detection range, and the measured value was outside the sensitivity range, resulting in inaccurate measurement data and loss of measurement value. Similarly, Ni et al. [20] used the sensor of nitrate to test various solutions with known concentration, and the results showed that the sodium nitrate sensor fluctuated greatly on the reading at high concentration. When studying the internet of things for aquaculture, Chen et al. [45] found that the dissolved oxygen sensor is prone to the probe damage failure caused by the pollution or the sensor being sank directly to the bottom of the pool. These failures pushed the sensor far away from the plant, and lead to the abnormal changes in the measured value of the dissolved oxygen sensor.

Sensors must be deployed in a variety of environments. As for the environmental impact, we need to pay attention to the design of a sensor in advance when deploying the sensor. The adaptive environment of the sensor should be fully considered during its deployment to avoid the sensor random failures due to environmental factors.

3. Sensors fault diagnosis

The earliest fault diagnosis method is to determine the cause and location of the fault after the fault occurs. The fault could be found with experience and simple methods. With the development of automation technologies, modern control systems become more complex and multi-functional. The previous methods could

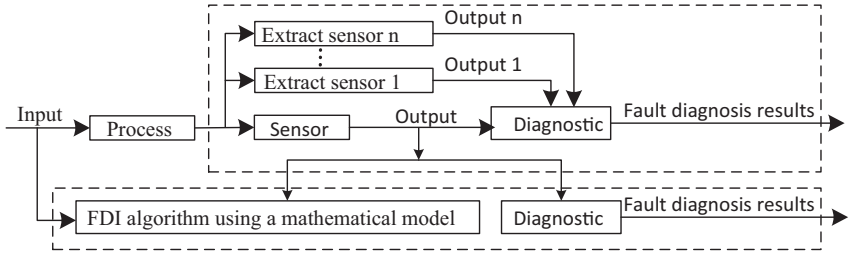

Fig. 2. The concepts of hard redundancy and analytical redundancy.

not meet the current requirements any more. Therefore, it is of great significance to study the fault detection and isolation methods [46]. With the wide applications of sensors in recent years, various sensor fault diagnosis techniques have been discussed [47].

In recent years, with the new development of fault diagnosis technology, there are many methods of sensor fault diagnosis from different perspectives. The concept of redundancy is one of the commonly used fault diagnosis methods. The method can be divided into hardware redundancy and analysis redundancy, as shown in Fig. 2. Hardware redundancy refers to measuring the same signal by using multiple sensor devices. A linear combination of measurements provided by redundant sensors is often used to obtain estimates and each measurement is compared to an estimate to determine if there is a fault. Redundancy analysis does not require additional hardware. Therefore, it is usually more cost effective than hardware redundancy methods.

The fault diagnosis task consists of three parts: fault detection, fault identification and fault estimation [48]. Firstly, the fault detection is to determine if the sensor is operating within the normal measurement range. Secondly, fault identification is to identify the source of the fault and identify what fault has occurred in the sensor. Finally, the fault level is determined by the fault estimation. In other words, it is to determine whether the sensor cannot continue to work, or it can temporarily maintain the working state.

3.1. Fault detection methods

Sensor fault diagnosis techniques are classified into model-based, knowledge-based and data-based [49].

3.1.1. Model-based approaches

In the model-based method, dynamic process models are applied to analyze the input and output signals. This method was first proposed by Willsky [50], but they did not give details of the processing. The difference between the signal of the model system and that of the actual system is defined as “the residual”. There is abundant fault information in the residual signal, and the fault can be diagnosed based on an appropriate decision function or decision rule. This is the earliest fault detection method, its principle is to detect whether the behavior in the model system is consistent with the behavior in the actual system. The method is often used together with the fault estimation method to construct the sensor fault diagnosis process. By establishing a model and choosing a decision function (or rule), the evaluation function of the residual is compared with the selected threshold function. As a result, a change is detected and a failure of the sensor system is judged [51]. This is an important and challenging method that is still popular today [52].

Arab et al. [53] proposed the modified parity space method, used parity space technology to generate linear analytical redundancy relations through individual observation of each sensor. The residuals are sensitive to sensor failures and obtained different differential relations. However, verification assumption is an ideal status that the sensor will fail. Hanen et al. [54] studied the fault detection and isolation of electric-drive sensor based by improving the parity space method. A fast and simple algorithm of sensor fault detection is designed for second-order systems. The simplicity of the final algorithm leads to the shorter execution time and less resource consumption in the conduction. The effectiveness of the method is verified by numerical simulation [55,56]. Hamed et al. [57] proposed a sensor fault detection method based on nonlinear parity technology, which can be used in pH neutralization system. The non-linear fault detection and recognition algorithm can effectively detect and isolate the sensor fault on the pH channel, and accurately detect the time of fault occurrence immediately. In general, the theoretical knowledge of this method is relatively mature and is mostly verified by simulation.

The latest model-based diagnosis method still has problems in sensitivity and practicability of algorithm detection. Many nonlinear dynamics of the actual working object can not be easily modelled. In many practical applications, it is difficult to obtain a model that can fully match the sensor behavior process [58]. However, the mismatch of the model may lead to large error signals [59]. These limitations make it difficult to develop and apply the fault detection method based on dynamic models of the real-time control system.

3.1.2. Knowledge-based approaches

In recent years, the development of computer technology has provided a new theoretical basis for fault diagnosis technology. It is difficult for many complex control systems to obtain accurate mathematical models due to the complexity of different fault types. Therefore, a fault diagnosis method based on knowledge is proposed. This method uses an expert system to locate and diagnose sensor faults. It does not need a quantitative mathematical model. It can analyze the fuzzy logic relationship and existing historical data in the reasoning engine of the system. Historical fault information and expert knowledge are used to diagnose whether a sensor has faults. This method has the ability of self-learning and high fault tolerance, and has advantages in the fault diagnosis of nonlinear systems. The structure of using expert systems for sensor fault detection is shown in Fig. 3.

The fault diagnosis method using expert systems is based on the fault information and knowledge of sensors. The overall system includes three main parts: knowledge base, rule base and reasoning mechanism. Knowledge rules and reasoning mechanism are used to analyze the working status of the sensor. The working process of the system is as follows: user extracts the measured value of the sensor that needs to be diagnosed, inputs a known fault phenomenon through the human-machine interface, and then the diagnostic system selects an appropriate inference step for the indeterminate reasoning based on the existing rules. In the process of reasoning, the system continues to interact with the user, asks if there are other faults, and finally identifies all possible faults [60]. According to the difference of inference machine, the knowledge-based sensors’ fault diagnosis method can be divided into rule-based diag-

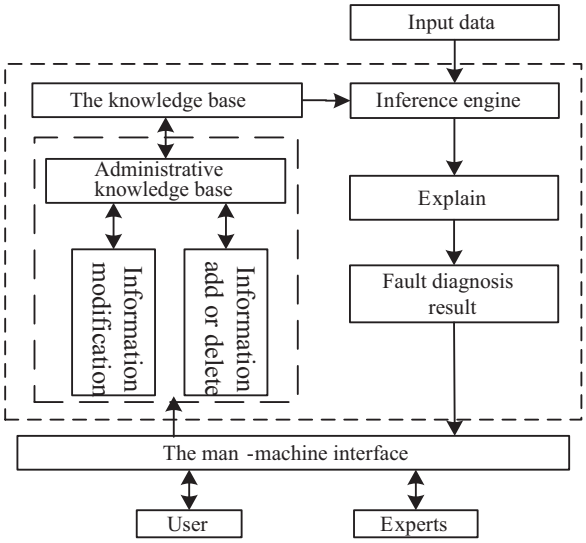

Fig. 3. The structure of expert systems for sensor fault detection.

nosis expert systems and fuzzy reasoning-based diagnosis expert systems.

4. Rule based intelligent system

Rule-based reasoning methods often rely on the experience of experts in the field as the basis for judgment.

Beaulah et al. [61] developed an expert system-based solution to diagnose, detect, and replace sensor fault readings. The inference section is designed to check the consistency and failure of the incoming signal. If everything is satisfactory, the data is passed to the optimization algorithm. If a fault is found, the inference module must be able to locate the fault, notify the user and derive the appropriate replacement value for the corrupted data. In the real-time environmental control of greenhouse microclimate, the effectiveness of its signal processing has been verified. Li et al. [62] proposed a new automatic fault detection system for aquaculture water quality monitoring devices. The system classifies the faults, and every fault object can obtain advice from experts. They also developed a web-based system interface where users can see the number of each type of failure that occurred during a particular period. Silva et al. [42] built an expert system by combining object-oriented modeling, rules and semantic network, and identified the four faults of sensor bias, drift, gain and signal loss through the expert system. The redundancy information of fault signal was extracted and calibrated.

The expression form of rule-based diagnostic knowledge is intuitional, and it can easily add or subtract rules to achieve fault diagnosis concordant. This is more efficient in solving small-scale problems. However, it is quite difficult to acquire rules for complex systems. Meanwhile, the reasoning process is directly related to the established rules. As a result, if the fault knowledge is not collected, then the system cannot detect it. Therefore, the system lacks adaptability to unknown problems.

5. Fuzzy inference based intelligent system

Fuzzy logic was initially proposed by Zadeh [63] as a tool to describe and process inaccuracy and fuzzy events. Subsequently, the development of fuzzy sets and special algorithms have solved many problems of control systems [64]. Unlike traditional logic systems, the purpose of fuzzy logic is to make rational decisions in the case of uncertain and inaccurate information. Fuzzy logic does not need the accurate mathematical model of the diagnostic object while uses function and fuzzy rule to make fuzzy diagnosis [65]. The method is robust in eliminating some of the limitations of early expert systems.

Kiupel et al. [66] replaced a crisp threshold for the fault decision with a fuzzy logic based threshold, and added natural experience and artificial common sense to the final decision in the diagnostic system. Pre-evaluating residuals using fuzzy logic prevents false alarms. In contrast to the conventional expert systems, the decision-making ability of this method is significantly improved. As systems become more complex, finding the cause of a failure is an extremely difficult task, because multiple failure modes can have similar and possibly indistinguishable patterns of effects. To locate sensors' malfunctions, Ouyang et al. [67] established a fuzzy expert system to locate sensor faults, and discussed the reliability of complex energy system components and sensor fault recognition diagnosis program. Geetha et al. [68] used the fuzzy logic to focus on residual generation and residual evaluation, and defined an intelligent fuzzy method. The experimental results indicate that the expert system isolates the sensor fault at the same instant of its occurrence, showing its instantaneous handling capability for the fault.

The method based on fuzzy expert systems reduces the dependence of rules on fault severity and load condition, lowers the cost and provides flexible design. However, it is difficult to determine the fuzzy relationship between faults and fault characteristics when dealing with actual cases. Moreover, when the fuzzy knowledge base is incomplete, it is easy to misdiagnose. Therefore, this method is now often applied together with other methods.

Knowledge-based fault diagnosis methods are far from perfect due to limited knowledge, single repeatability of rules, and subjectivity of function definition.

5.0.1. Data-driven approaches

The method of data-driven fault diagnosis is an emerging method. Mehrdad et al. [69] extended a model-based method to data-driven method. It does not need quantitative mathematical model, but applies process data directly. This means that hidden information in the data can be collected by processing and analyzing large amounts of data. In this way, the fault can be easily found [70]. The problem of sensor fault recognition is considered as pattern recognition in this method. Based on data processing, the sample data is acquired and trained to obtain a classifier, and then matches the data according to the classification rules [71].

6. Neural network

With the development of efficient neural network and learning algorithm, system knowledge and diagnosis knowledge are integrated into neural network diagnosis algorithm [72–74]. A neural network is a massively parallel adaptive network composed of simple nonlinear computing elements. They are designed to abstract and simulate certain functions of the human nervous system, and these algorithms are often used to identify faults based on extracted features [75].

The sensor fault diagnosis method based on neural network mainly utilizes the nonlinear mapping ability of neural network to realize the nonlinear conversion from input space to output space. It also has the ability of memory and speculation to process problems in parallel [76]. In the sensor faults detection process based on neural network, the network structure and the appropriate activation function is determined according to the characteristics of the output vector dimension and the number of faults. The network is trained using the fault samples. By repeatedly adjusting

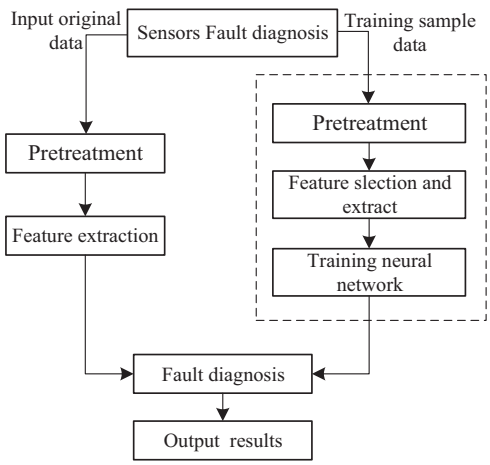

Fig. 4. Sensor fault detection based on neural network.

the learning rate, inertia factor and other parameters, the network continuously improves until it reaches the ultimate precision [77]. The detection process is shown in Fig. 4. The research of typical neural networks is mature and used widely. At present, the research on neural network diagnosis methods mainly focus on the improvement of the training data algorithm [78].

Sensors' fault diagnosis methods based on neural network are often used in the complex systems. In order to detect sensor failure on the combine harvester, Geert et al. [79] introduced the artificial neural network into the combine harvester to diagnosis sensor fault. However, the data used in this case is static, which can lead to significant time delays in the fault isolation procedure. Ferentino et al. [80] adopted the back-propagation training algorithm to train data. Since the training process adopted is offline, they can overcome the influence of plant defects on the training accuracy and isolate the information features from specific sensors. Their experiments proved the feasibility of online fault detection based on neural networks. Balaban et al. [81] utilized complex aerospace brakes containing temperature, vibration, current, and position sensors, and they configured the most common sensor failures (such as bias and noise). They injected these faults into the experimental data and established a fault classifier based on neural networks to identify the types of faults. The method based on neural networks has been proved to be able to better implement the health management of sensors in complex systems [82]. UAVs are now also a fast-growing area. Samy et al. [83] used extended minimum resource allocating radial basis function (EMRAN-NN) based on NN scheme to model the continuous failure of sensors in UAV. The method is proved feasible by using actual data. However, in the experiment of fault-tolerant flight control systems, some fault accommodation errors are quite high which must be reduced by further research. Imran et al. [84] found that it was necessary to strictly control the short-term storage temperature of flour, and proposed a temperature sensor fault detection method based on online learning. They used a feedback artificial neural network trained by a random sequence-learning algorithm to detect abnormal measurements of physical sensors. Meanwhile, they also used auxiliary sensors to replace the failed main sensor with redundant analysis and continue to work to increase system efficiency.

The literature shows that the neural network has advantages of being able to work in real time and it can use time sequence data directly. For now, this is a widely used method. However, neural networks suffer from inherent defects, such as the need for a large number of training samples, and poor generalization capabilities. These limitations will increase the difficulty of fault detection.

7. Support vector machines

This is a classification method based on limited sample information [85]. In the fault diagnosis process, the normal data is different from the fault data, and the fault data belongs to the abnormal class. By training the sample data repeatedly for grasping the physical laws of the system, it can find the deviation from the normal model during the sensors' working process, and this deviation will be recognized as ãbnormality. A diagnostic method based on support vector machine (SVM) can detect multiple faults [86,87]. The process of SVM-based fault diagnosis method is shown in Fig. 5.

Zhu et al. [88] proposed a sensor fault diagnosis and prediction model based on SVM in order to overcome the shortcomings of the neural network model, which is prone to fall into the local minimum and has a strong dependence on the number of practical samples. Compared to the traditional methods, it is much more effective and accurate. Deng et al. [89] proposed the sparse least squares SVM (OS-LSSVM) as the fault diagnosis method for sensor fault detection and recognition. By changing the threshold, the number of support vectors is reduced, the dimension is reduced, and the operation speed is improved. Liu et al. [4] proposed a fault diagnosis method of water quality sensor based on multi-stage SVM. Firstly, they used domain knowledge to construct a decision tree to help remove irrelevant data. Four different MSVM kernel functions of SVM model are compared to classify sensor data. The results show that the accuracy of RBF kernel model is 92.86 %.

In the case of limited fault information, SVM can mine the implicit classification information to the maximum extent and classify fault features into specific fault categories [16]. It does not need to know the prior knowledge of the physical reasons of the controlled system and to establish the corresponding fault model in advance. This overcomes some empirical problems such as parameter selection sensitivity to some extent [86,90]. Compared with traditional classifiers, SVM classifiers have the better generalization performance. When training support vector machine classifiers, the risk of structural classification errors is minimized, while tra-

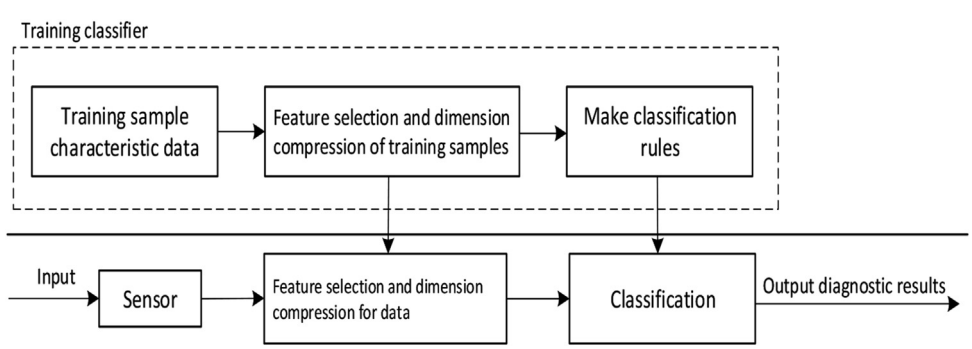

Fig. 5. Process of SVM-based fault diagnosis method.

ditional classifiers usually minimize the experience risk through training. This method can use a small number of samples to solve learning problems, but the classification accuracy of SVM depends on the complexity of the data.

SVM is the main research direction due to its excellent classification performance, but this method requires more labeled samples and suffers from data imbalance, which will lead to higher costs [91].

8. Deep learning

Deep learning is a particularly complex neural network and has been developed rapidly [92]. It is a relatively new fault diagnosis algorithm, which solves issues related to poor generalization ability of shallow neural network and insufficient characterization ability of complex functions. Moreover, it is capable of automatic feature extraction, which avoids manual feature extraction process and enhances fault diagnosis. It has significantly improved the accuracy of recognition of many traditional recognition tasks and its theory and performance have received a significant amount of attention from researchers. At present, this method is widely used in image processing, natural language processing, etc.

As an emerging field, the current application in sensor fault diagnosis also shows its powerful function. However, there is still a lack of research on the use of deep learning methods to detect sensor faults. Lee et al. [93] proposed a deep learning algorithm. This method can directly extract the decisive features of fault detection from the sensor signals and use these features to classify the main faults. Mandal et al. [94] used deep belief network (DBN) technology to classify the fault data. Their method adopted layer-by-layer unsupervised preprocessing to improve classification accuracy.

In recent years, data-driven methods have gradually emerged in fault diagnosis process systems. These methods can be introduced for the sensor fault detection. Kruger et al. [95] proposed an online fault diagnosis method based on reconstruction based contribution (RBC), which can identify process measurements that are closely related to triggering alarm. Compared with the original monitoring system, it is found that the data-driven wind energy conversion system (WEC) monitoring system can trigger fault alarms in advance. Jiang et al. [96] developed a data-based key performance indicator fault detection toolbox (DB-KIT). The toolbox can be obtained free of charge online for people to detect faults. In [97], the research progress of data-driven methods over the past two decades has been reviewed. A new data-driven fault detection method is proposed for industrial cyber physical systems.

Data-driven methods are attractive, because it is mainly based on a large amount of historical data, and it is not necessary to exact knowledge of the process parameters. It is the more popular detection method.

8.1. Sensor fault identification methods

Fault identification determines the size and time-varying nature of the fault [98]. By analyzing the response characteristics of the signal amplitude, phase, and spectrum, many distinct or implicit interconnections can be used to identify faults in sensor system. The boundary between fault and health status can be described by diagnostic index without precise modeling of diagnostic object [99].

8.1.1. Principal component analysis

Principal component analysis (PCA)-based fault identification has been widely used in sensor fault identification. PCA is a statistical analysis method and can be used to find factors with much lower dimensions than the original data set so that the major trends in the original data set can be properly described. When the sensor fails, the PCA model will be destroyed, and the data can be analyzed to reveal the nature of the fault for identification [100].

Zhao et al. [101] reported that sensor failures accounted for more than 80 % of the total failure in the aviation system. They used PCA to establish a diagnostic model for each status and proposed division-based sensor fault diagnosis method in flight status. The experimental results show that it can effectively promote the rapid identification of single fault sensors. Shen et al. [102] designed a relevance vector machine predictor to evaluate the system, and they used PCA to detect multifunctional self-validating sensor faults. Wang and Liu et al. [103] observed the real-time monitoring sensor of food storage and used PCA to model the measurement data by dividing the measurement space into molecular space and residual subspace. They used the square prediction error and sensor validity index for fault identification.

Sensors in agriculture are often placed in harsh environments with high humidity and temperatures. In order to ensure the accuracy of sensor data of measurement and control system in a greenhouse environment, Wang et al. [104] proposed a fault identification method based on PCA. They used of the temporal and spatial correlation characteristics of environmental information. They supplemented the spatial similarity comparison nodes with historical data in short-term sensors, and used adjacent nodes to predict data for fault location. Compared with the traditional method, this method improves the detection accuracy by 22 % and reduces the false alarm rate by 15.7 %. The main disadvantage of PCA monitoring approaches is that PCA is a linear method whereas most real systems are nonlinear. Therefore, modelling a nonlinear system by applying PCA method as the linear solution can lead to fault diagnosis performance degradation. Kernel principal component analysis (KPCA) is a new nonlinear principal component analysis technique for nonlinear systems [105]. Mania et al. [106] improved PCA and proposed a new method of nuclear principal component analysis (KPCA). They tested the sensors in the underwater vehicle. The results show that compared with the PCA, the KPCA method has higher sensitivity to faults and can be more effective for producing warning signals.

This method does not need to establish a precise mathematical model, and it can transform high-dimensional data into low-dimensional data.

8.1.2. Wavelet transform

Wavelet transform (WT) can locate the signal characteristics of the sensor accurately in both time and frequency domain. It is suitable for non-stationary signal analysis and has been widely used in the fields such as signal processing and fault identification [107]. Catastrophe points and singular points of signals contain a large amount of important information that can reflect the inherent characteristics of the signal source or abnormalities or malfunctions of certain parts of the signal.

When using WT based sensor fault diagnosis, the algorithm is used in multi-level wavelet decomposition of the signal to obtain each sub-band data [108]. By applying wavelet transformation on the signal, the limit point caused by the input can be eliminated, and the remaining limit points can be defined as the fault signal. The wavelet transform can separate the fault signal effectively and display it on the time spectrum clearly. Wavelet transform can be divided into continuous wavelet transform and discrete wavelet transform. Continuous wavelet transform can be used to distinguish signal catastrophe points and noise; discrete wavelets transform can be used to detect the frequency structure of a mutation, signal mutation, or singularity. The process of sensor fault localization by wavelet transform is shown in Fig. 6.

Zhang et al. [109] introduced a new sensor fault diagnosis algorithm based on wavelet transform. The method decomposes the output signal of the sensor into elementary building blocks of

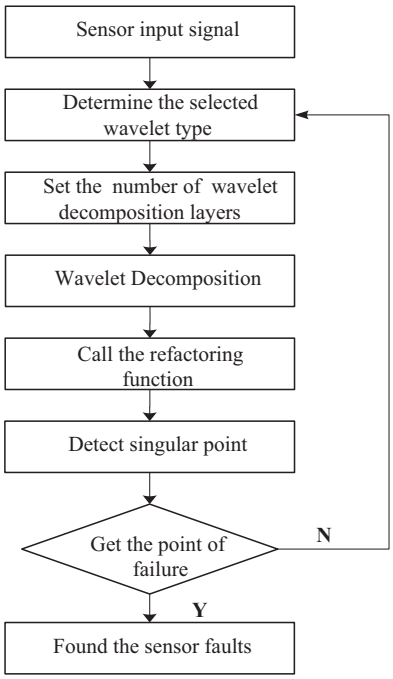

Fig. 6. The process of sensor fault localization by wavelet transform.

wavelet transform to detect local signal changes caused by sensor faults. The abnormal status of all sensors will introduce some transient changes in the output. The local information obtained by wavelet transform can identify the moment when the sensor is in an abnormal status, and classify the fault by using the instantaneous signals before and after the difference in signal distribution on each decomposition wavelet scale. Alexey et al. [110] reported a method based on wavelet to detect the fault of sensors. However, this method only detects the sudden fault of the sensor, and it is difficult to find the slowly changing fault. In view of the noise effect, Hou et al. [111] proposed a sensor fault diagnosis method based on mathematical morphology. They combined a modified generalized morphological filter with multi-scale wavelet transform to provide a new fault detection and diagnosis method for sensor grading and mutation. Li et al. [112] used the coarse scale and detail scale based on wavelet multi-resolution analysis to detect the drift fault and the progressive degradation fault. In order to improve the reliability of sensors in AUV, Wang et al. [113] studied non-visual sensors without their own fault diagnosis system in AUV. They combined wavelet transform with linear smoothing algorithm and its application in Shandong, China, showed a satisfactory results.

The WT-based identification method is fast and simple. It has high diagnostic accuracy, significant diagnostic effect, lower system model dependence and good localized characteristics. It is suitable for extracting mutant components from slow-changing signals. However, this method is only suitable for analyzing non-stationary signals in cases of sudden changes in the frequency structure of the system [114]. Therefore, this method can only effectively identify the sudden failure of sensors. In addition, when the wavelet scale is small, there will be jitter near the fault point, which cannot be used for accurate fault detection. When the wavelet scale is large, there will be large detection delay, which will affect the use of the sensor [115].

8.2. Sensors fault estimation methods

Sensor fault estimation provides information about fault severity, frequency and time characteristics by measuring the relationship between signals and output signals [116].

As the residual contains fault information, it is the basis for the fault estimation. The process of generating residuals can be summarized in two steps (as shown in Fig. 7): firstly, it is necessary to establish a mathematical model of the system to be tested and determine the corresponding model parameters. The second step is to establish the system fault model, which is based on the possible fault of the system and generate the residual error. These residuals should be zero if the system is working properly [117]. The residual is the basis of fault diagnosis. Fault estimation is often used together with model-based fault detection methods.

According to the generation mode of residual error, it can be subdivided into the parameter estimation diagnosis method and the state estimation diagnosis method [118].

8.2.1. Parameter estimation

The parameter estimation approach employs the mathematical model of online identification in order to determine the physical coefficients of the process. Sensor components have their own unique physical parameters, which are used to describe the local behavior of sensor components. It detects the fault according to the statistical characteristics of the model parameters and its corresponding physical parameters. This detection method can directly establish the model. Firstly, the working mechanism of the system is analyzed, and the relationship between model parameters and output is obtained. Then, the real value of the measured data is obtained online. The system calculates the output, and compare the true output and the model parameter output [119]. When the system parameter estimate exceeds a predetermined confidence interval, the fault is characterized by a significant change in the parameter. The nonlinear system sensor fault diagnosis filter based on parameter estimation has poor stability and low estimation accuracy. Parameter estimation is applied offline to obtain a discrete model. Weber et al. [120] used a Page-Hinkley algorithm to detect faults. The parameters of several output error models of the residual are estimated.

8.2.2. State estimation

The state estimation method is also called the filter method. The basic idea is to use the measurable information of the system to establish an analysis model, designs a system observer (detection filter), and compares the estimated value and the measured value to estimate and separate the sensor faults. The observer based technique has been widely used in estimating and reconstructing measurement noise and sensor fault [121,122].

Gao et al. [123] proposed a novel fuzzy state / disturbance observer for the augmented fuzzy descriptor systems, and obtained simultaneous estimates of the sensor's original state and disturbance state. Sanjay et al. [124,125] proposed a method for sensor fault detection and isolation in an extended Kalman filter structure. They conducted simulation experiments and introduced an error measurement to identify the fault sensor. This method further improved the estimation performance of Kalman filtering in nonlinear systems [126]. Sliding mode observers have the advantages of simplicity, robustness and high reliability.

Several researchers exploited the sliding mode observer to solve problem of fault reconstruction and the estimation of robust sensors [127]. Du et al. [128] proposed an approach of fault-tolerant control for the reconfigurable manipulator fault based on sliding mode observer. This method makes the diagnostic system remain stable in the presence of unknown faults and modeling uncertainties. Liu et al. [129] proposed a new descriptor sliding mode approach that can obtain accurate estimates of system states, sensor fault vectors, and disturbances simultaneously. This scheme for integral sliding mode control can stabilize the fault system. Rios et al. [130] proposed a method combining the high-order sliding mode multi-observer technology and multi-model technology.

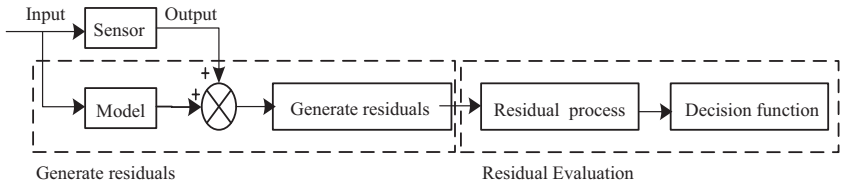

Fig. 7. Sensors fault estimation process.

This method has the advantages of both a sliding mode observer and multiple models. Halim et al. [131] pioneered a sliding mode observer for sensor fault estimation. This method is achieved by proper scaling and filtering of the so-called equivalent output error injection, which represents the average value. Chen et al. [132] proposed a fault detection method based on strong tracking filter and first-order time correlation function model for fault diagnosis and signal recovery of speed sensor in autonomous underwater vehicles (AUV). They obtained real experimental data of a series of AUV trajectories by applying this algorithm to AUV trajectory. The results showed that this method could diagnose common faults in time, and give signal recovery for a period. This method can guarantee the stability of AUVs. Jmal et al. [133] proposed a robust sensor fault estimation method for systems with monotonic nonlinearity and unknown inputs. Jiang et al. [134] designed a new observer to perform fault detection for dynamic systems. They proposed a multiple-input multiple-output (MIMO) algorithm that uses a residual generator instead of multiple residual generators. This approach reduces the design effort and implementation complexity of the fault detection systems.

This method uses simple algorithm and has fast response speed. At present, a variety of mature sensor fault estimation methods are developed, while some of them are combined with other model-based methods.

8.3. Hybrid fault diagnosis approach

Effective fault diagnosis of sensors requires the combination of detection, identification and estimation techniques, but each technique has its own advantages and disadvantages. If applied reasonably and effectively together, they can complement each other, and detect faults more timely and effectively.

Xiao et al. [135] proposed a method for dealing with fault detection and diagnosis in wastewater treatment process using neural network and ARMA model. Yoon et al. [136] proposed a model-free sensor fault detection and isolation technology based on the combination of the parity equation approach and discrete wavelet transform, which can detect and isolate multiple faults of finite redundant systems. In addition, this hybrid technique does not need any dynamic model except the sensor arrangement information, so it is more efficient than the analytic redundancy method. Mehrdad et.al [69] used a least-square support vector machine (LS-SVM) to generate system residuals, which were analyzed by discrete WT to detect the sensor fault and plant fault. The performance highly relies on the accuracy of the predictor model. Hadad et al. [137] proposed a PCA-based neural network to improve the limitations of neural network detection methods. Compared to traditional neural networks, this type of neural network is small and easy to train. Principal component analysis is used to reduce the input vector and reduce the input matrix. By reducing the problem dimension, the response accuracy, efficiency and robustness of the neural network to the noise data are significantly improved. Zhang et al. [138] proposed an effective method combining principal component analysis and neural network for fault sensor detection in multiple sensors. It detects and identifies sensor faults in transient condition. It is better than traditional PCA in the identification of deviation and drift fault.

Ni et al. [139] proposed a detection method based on the combination of kernel principal component analysis and support vector machine. This method introduced the trust mechanism and improved the adaptability of the KPCA model to emergency monitoring. In addition, the method can accurately distinguish between two emergencies. Similarly, Shen et al. [102] combined the two methods of relevance vector machine (RVM) and principal component analysis (PCA), and proposed a new method of self-verifying sensor fault detection, isolation and recovery. It can effectively isolate multiple concurrent faults of sensors and has high precision. Compared with the traditional neural network, the time consumed is less and the recovery error is smaller. Both neural networks and fuzzy systems are important methods for soft computing. Neural networks can process information in parallel, and its self-learning ability is strong, but the knowledge form expressed by weight is not easy to understand. Fuzzy logic can make good use of language knowledge, and the expression form is easy to understand. Its shortcoming is that it is weak in self-learning ability, and it is difficult to use numerical information. Therefore, some researchers using a combination of the two form fuzzy neural networks. This fuzzy system is expressed as a fuzzy neural network. This method reduces the amount of training parameters, and the diagnosis of multi-site faults can be performed online at the same time [140]. Carlos et al. [141] designed a fault-tolerant system for dissolved oxygen sensors in sludge that are prone to bias and drift faults. The core idea is to combine adaptive neural network with fuzzy inference system through fuzzy logic to perform fault classification. In order to achieve fault estimation, an artificial neural network method is adopted. The simulation results show that the method responds faster to slow drift fault types. Kwon et al. [142] proposed a sensor estimation method based on fuzzy logic and neural network. Traditional fault diagnosis of sensors in unmanned aerial vehicles is based on hardware redundancy, and multi-group measurement data is usually used for comparison. Nevertheless, this method will add extra weight to the drone that will increase the cost. The analysis of redundancy is usually based on the model, and no additional sensors are required. However, the mathematical model established by the analysis of redundancy may be different from the actual system, and unknown interference will affect the system. Combining the two methods can address these shortcomings.

Selecting the appropriate detection, identification, and estimation methods, and combining them, can more quickly identify sensor failures. This diagnostic method is commonly used at present.

9. Conclusion

This paper reviews the development status of the sensors' fault diagnosis technologies and its significance in industry and agriculture. It summarizes the common fault types of sensors in monitoring and control systems. Future development of sensors is expected to ensure sensors to work not only with higher efficiency,

but also with greater reliability. The safe and reliable operation of sensors depends on the following points:

- The manufacturing technology of sensors needs to improve sensors' quality to make them less prone to faults and more adaptable to the complex environment;
- A comprehensive sensors' fault diagnosis system should be configured, which can detect the fault as soon a small error is occurred, so as to prevent the complete failure of the sensor.

The review demonstrates that the latest fault diagnosis methods have combined different advanced technologies including intelligent systems, support vector machines and deep learning. These advanced technologies aim to address the main challenge of reducing the computational complexity of the diagnosis system. Diagnostic algorithms should have low complexity and be able to immediately detect and diagnose fault signals. In addition, it should be able to distinguish different types of faults and identify the occurrence of a new fault without classifying it as one of the known faults or standard operation. The most promising methods is to use deep learning and big data analytics. In addition, some of the latest FDI technologies in other areas (such as fault diagnosis methods applied to system processes) can be applied for effective detection of sensor faults.

Future research should address the challenges in developing fault tolerant control of sensors and the compensation measures for avoiding the sudden sensor failure.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.sna.2020.111990.

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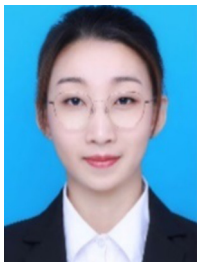

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